HomeSphere - Smart Home Services Platform

Project Description for Hackathon Submission

1. Project Overview

HomeSphere is a comprehensive web-based platform designed to bridge the gap between households needing daily services and local service professionals. By leveraging modern web technologies and real-time data, HomeSphere simplifies the process of finding, booking, and managing home services like cleaning, repairs, plumbing, and electrical work.

2. Problem Statement

In the current market, finding reliable local help is fragmented and often unreliable.

-For Users: No centralized platform to find verified professionals, track work progress, or manage payments transparency.

- **For Helpers:** Struggle to find consistent work, manage their schedules, and track their earnings.

- **For Administrators:** Lack of data-driven insights to manage service quality and platform growth.

3. Solution

HomeSphere provides a unified ecosystem with three distinct roles:

- **User Portal:** For booking services and tracking status.

- **Helper Portal:** For accepting jobs and managing workflow.

- **Admin Dashboard:** For platform oversight, analytics, and user management.

4. Key Features

🛡️ Admin Dashboard (The "Command Center")

- **Real-Time Analytics:** Visualizes platform health with dynamic charts (powered by **Chart.js**) showing revenue trends and service distribution.

- **Live KPIs:** Tracks Total Bookings, Active Users, Total Revenue, and Cancellation Rates in real-time.

- **Booking Management:** comprehensive view of recent bookings with status filtering and sorting.

- **Data-Driven Insights:** Helps administrators make informed decisions based on service demand and financial performance.

👤 User Experience

- **Seamless Booking:** Easy-to-use interface to select services (Cleaning, Repair, etc.).

- **Real-Time Tracking:** Users can see the status of their service request (Pending -> Assigned -> In Progress -> Completed).

- **Service History:** Access to past bookings and payment details.

👷 Helper Tools

- **Job Board:** View available service requests in the vicinity.

- **Work Management:** Simple tools to update job status (e.g., mark as "In Progress" or "Completed").

- **Earnings Tracker:** Transparency in payments and completed jobs.

5. Technology Stack

- **Frontend:** HTML5, **Tailwind CSS** (for modern, responsive, and glassmorphism design).

- **Logic & Interactivity:** Vanilla JavaScript (ES6+).

- **Backend-as-a-Service: Firebase**(Google).

  - **Firebase Authentication:** Secure email/password login and role-based access control.

  - **Cloud Firestore:** Real-time NoSQL database for syncing data across users, helpers, and admins instantly.

- **Data Visualization: Chart.js** for rendering interactive graphs and charts.

- **Icons:** FontAwesome.

6. Innovation & USP

- **Role-Based Access Control (RBAC):** Secure and distinct experiences for Admins, Users, and Helpers within a single application.

- **Instant Synchronization:** Leveraging Firestore, updates (like a helper accepting a job) are reflected instantly across the platform without page reloads.

- **Scalable Architecture:** Built on serverless Firebase infrastructure, capable of handling thousands of concurrent users.

7. Future Scope

- **AI Integration:** Implementing ML models to predict service demand and optimize helper allocation.

- **In-App Payments:** Integration with payment gateways (Stripe/Razorpay) for secure transactions.

- **Mobile App:** Developing a React Native version for on-the-go access.

- **Chat System:** Real-time chat between users and helpers for better coordination.

Team Name: TeamXCoder

Project Link: https://github.com/Adityaguptaaa/Homesphere